

Report of Existing Systems

Existing Systems

Web-Based Systems

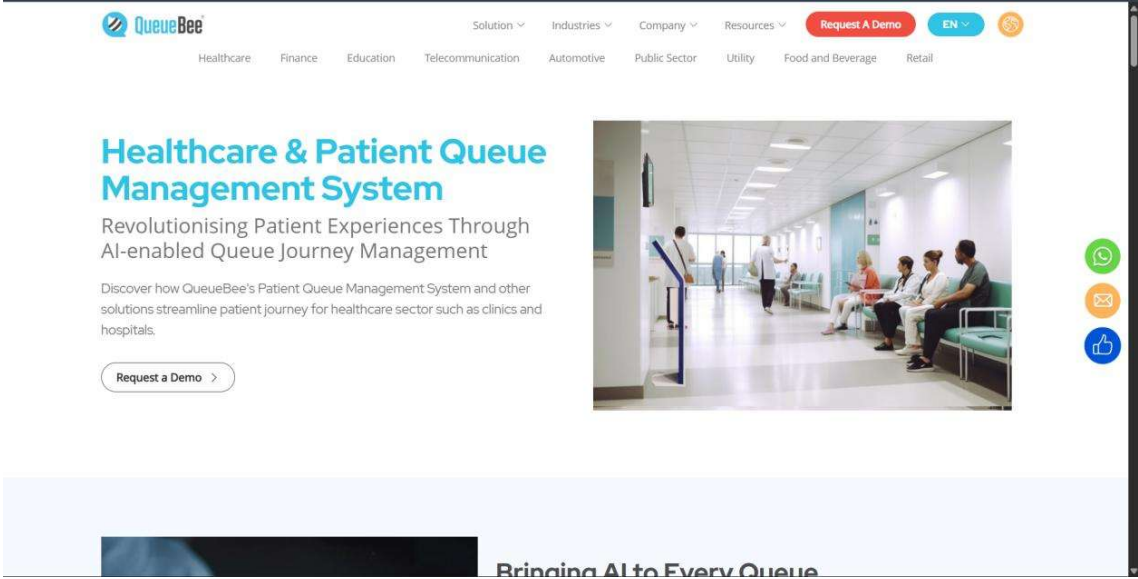
- Queueasy

The screenshot shows the Queueasy website with a blue header. The main content area features the headline "Transform Hospital Patient Experience" and a sub-headline "Streamline OPD, pharmacy, billing, and lab queues with our intelligent token system. Reduce wait times by 65% and boost patient satisfaction with real-time queue management." A live update box states "Queues processing 2.5x faster" with a note that the average wait time has been reduced from 47 to 18 minutes. A "Live Queue Status" widget displays "Current Patients: 12" and "Next Token: #023". A "PATIENT SATISFACTION" badge shows "87%" with a note "Increased by 42%". The footer includes a "Book Free Demo" button and a "Watch Demo" button.

- Robato Smart Queue Management System

The screenshot shows the Robato Systems website with a dark blue header. The main content area features the headline "Smart Queue Management System" and a sub-headline "Streamline your customer flow and enhance service efficiency with our intelligent queue management solution." A "Welcome to Robato Systems" message box contains a form with fields for "Name", "Email", "Contact Number", and "Interest". A "Free Demo" button is visible in the top right. The footer includes a "Queue & People Management System" button and an "Online" button.

- QueueBee



QueueBee

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Healthcare Finance Education Telecommunication Automotive Public Sector Utility Food and Beverage Retail

Healthcare & Patient Queue Management System

Revolutionising Patient Experiences Through AI-enabled Queue Journey Management

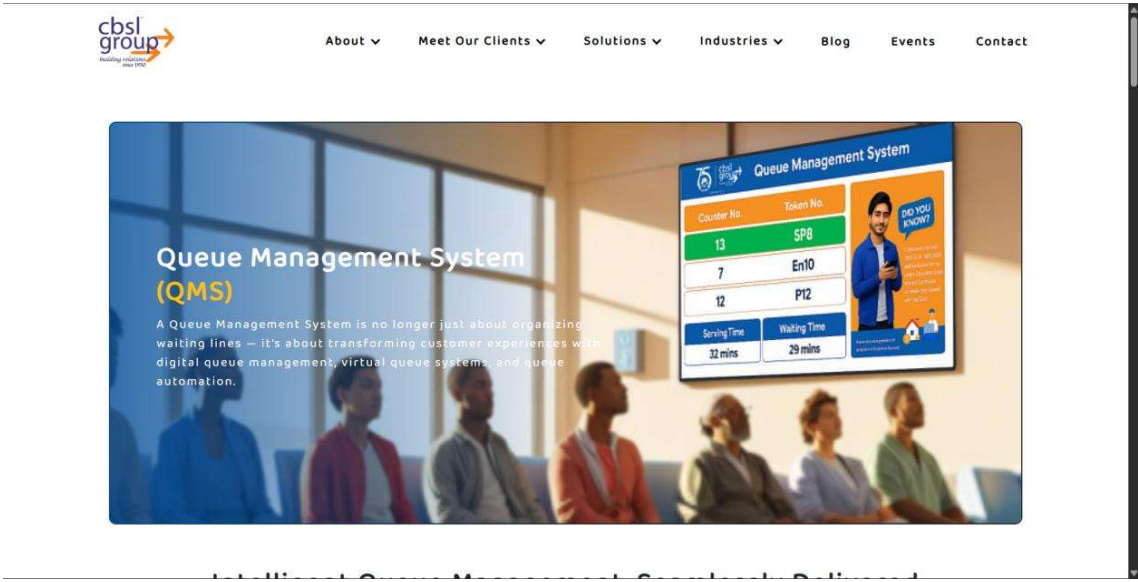
Discover how QueueBee's Patient Queue Management System and other solutions streamline patient journey for healthcare sector such as clinics and hospitals.

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Bringing AI to Every Queue

- CBSL Queue Management System



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Queue Management System (QMS)

A Queue Management System is no longer just about organizing waiting lines – it's about transforming customer experiences with digital queue management, virtual queue systems, and queue automation.

Counter No.	Token No.
13	SP8
7	En10
12	P12

Serving Time	Waiting Time
32 mins	29 mins

DO YOU KNOW?
Queue management system helps you to manage the queue of your customers and provide them with the best service.

Intelligent Queue Management System for Banks

Mobile Applications

- QLess

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QLess - Queuing Software

QLess, Inc.
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About this app

Common Features in Existing Systems

Feature
1. User Registration/Login
2. Appointment Booking
3. Digital Token Generation
4. Queue Status Tracking
5. SMS/Push Notifications
6. Queue Display Dashboard
7. Admin Management
8. Basic Analytics
9. Queue Cancellation/Rescheduling
10. Prediction Dashboard

Limitations of Existing Systems

Limitation	Description
1. No Accurate AI Prediction	Most systems estimate wait times using simple calculations.
2. No Smart Slot Recommendation	Users must manually choose available slots.
3. Limited Queue Optimization	Queues are generally first-come-first-served.
4. Poor Adaptation to Real-Time Changes	Sudden delays can make estimates inaccurate.
5. Limited Resource Forecasting	Cannot predict future workload effectively.
6. Lack of Multi-Domain Support	Often designed for only hospitals or only banks.
7. Minimal IoT Integration	Few systems use sensors for automatic crowd monitoring.

Proposed System

AI-Based Smart Queue Prediction and Intelligent Reservation System (IntelliQ)

The proposed system enhances traditional queue management by using Artificial Intelligence, Machine Learning, and Real-Time Analytics to predict waiting times and recommend optimal reservation slots.

Unique Features of IntelliQ

Feature	Existing Systems	IntelliQ
1. Appointment Booking	Yes	Yes
2. Digital Token System	Yes	Yes
3. Real-Time Queue Tracking	Yes	Yes
4. Notifications	Yes	Yes
5. AI Waiting Time Prediction	No	Yes
6. Smart Slot Recommendation	No	Yes
7. Peak Hour Forecasting	No	Yes
8. Dynamic Queue Optimization	No	Yes
9. Resource Utilization Prediction	No	Yes
10. Multi-Sector Support	Limited	Yes

Feature	Existing Systems	IntelliQ
11. IoT-Based Crowd Monitoring (Optional)	Rare	Yes
12. Gesture-Controlled Kiosk (Optional)	No	Yes
13. Predictive Analytics Dashboard	Limited	Yes

Comparative Analysis

Criteria	Existing Systems	IntelliQ
1. Queue Management	Basic	Intelligent
2. Waiting Time Estimation	Approximate	AI-Based Prediction
3. Reservation System	Static	Smart Recommendation
4. Adaptability	Low	High
5. Automation Level	Medium	High
6. User Experience	Good	Enhanced
7. Operational Efficiency	Moderate	High
8. Scalability	Limited	High

SRS Report of IntelliQ

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Chapter 1: Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the **AI-Based Smart Queue Prediction and Intelligent Reservation System (IntelliQ)**. This document provides a detailed description of the system's objectives, features, architecture, constraints, and user requirements. It serves as a reference for developers, testers, project supervisors, and stakeholders throughout the software development life cycle.

The proposed system aims to modernize conventional queue management by integrating Artificial Intelligence (AI), Machine Learning (ML), cloud computing, and real-time analytics to predict waiting times, recommend optimal appointment slots, and improve operational efficiency across multiple service sectors.

1.2 Scope

The **AI-Based Smart Queue Prediction and Intelligent Reservation System (IntelliQ)** is a cloud-based, cross-platform application designed to simplify appointment booking and queue management for both users and service providers.

The system enables users to register, book appointments, receive digital queue tokens, monitor queue status in real time, and receive AI-based waiting time predictions along with smart slot recommendations.

Administrators can efficiently manage services, appointments, counters, and queue operations while accessing analytical reports and predictive dashboards to support better decision-making.

The proposed system is intended for multiple sectors, including:

- Hospitals and Healthcare Centers
- Banks and Financial Institutions
- Government Offices
- Educational Institutions
- Public Service Centers
- Customer Service Organizations

The system is designed to support Android, iOS, Windows, macOS, Linux, and Web platforms using a single Flutter codebase with a centralized cloud-based backend.

1.3 Objectives

The primary objectives of the proposed system are:

- To reduce customer waiting time through intelligent queue management.
- To provide an online appointment booking and reservation system.
- To generate digital queue tokens automatically.
- To provide real-time queue status tracking.
- To predict waiting times using Artificial Intelligence and Machine Learning techniques.
- To recommend the most suitable appointment slots based on historical and real-time queue data.
- To forecast peak hours and improve resource utilization.
- To provide administrators with analytical reports and predictive dashboards.
- To support multiple organizations using a centralized cloud-based platform.
- To improve customer satisfaction through notifications and intelligent scheduling.
- To provide a scalable and secure solution that supports future IoT and AI enhancements.

1.4 Technologies Used

Component	Technology
Frontend	Flutter
Programming Language	Dart
Backend	ASP.NET Core Web API
Database	Azure SQL Database
ORM	Entity Framework Core
AI / Machine Learning	Python, Scikit-learn, Pandas, NumPy
Real-Time Communication	SignalR
Authentication	JWT Authentication
Push Notifications	Firebase Cloud Messaging (FCM)
API Testing	Postman
Version Control	Git & GitHub
IDE	Visual Studio 2026, Visual Studio Code, Android Studio

Chapter 2: Overall Description

2.1 Product Perspective

The AI-Based Smart Queue Prediction and Intelligent Reservation System (IntelliQ) is a cloud-based intelligent queue management platform designed to improve traditional appointment and queue handling systems.

The system follows a **three-tier client-server architecture**, where Flutter applications act as clients, ASP.NET Core Web API serves as the application layer, and Azure SQL Database stores all organizational and user data. AI prediction services analyze historical and real-time queue data to estimate waiting times and recommend optimal appointment slots.

Future versions may integrate CCTV-based crowd monitoring and IoT devices to improve prediction accuracy and automate crowd analysis.

2.2 Product Functions

The proposed system provides the following major functions:

- User Registration and Authentication
- Appointment Booking
- Digital Token Generation
- Real-Time Queue Tracking
- Queue Cancellation and Rescheduling
- Push Notifications and Alerts
- AI-Based Waiting Time Prediction
- Smart Slot Recommendation
- Peak Hour Forecasting
- Dynamic Queue Optimization
- Resource Utilization Prediction
- Predictive Analytics Dashboard
- Organization and Service Management
- Multi-Sector Support
- Optional CCTV/IoT Crowd Monitoring
- Optional Gesture-Controlled Kiosk

2.3 User Roles

The proposed system consists of multiple user roles, each with specific responsibilities and access permissions. Role-Based Access Control (RBAC) is implemented to ensure secure and authorized access to system functionalities.

2.3.1 User (Customer)

The User is the primary consumer of the system who books appointments and tracks queue status.

Responsibilities:

- Register and log in to the system.
- View available organizations and services.
- Book appointments by selecting preferred date and time.

- Receive digital queue tokens.
- View real-time queue status and estimated waiting time.
- Receive AI-based slot recommendations.
- Cancel or reschedule appointments.
- Receive push notifications and reminders.
- View appointment history and profile information.

2.3.2 Counter Staff / Operator

Counter Staff are responsible for serving customers and managing the queue at their assigned service counter.

Responsibilities:

- Log in to the system.
- View assigned service counter.
- Call the next customer in the queue.
- Mark services as completed.
- Skip absent customers.
- Pause or resume queue operations when required.
- Update service completion status.

2.3.3 Super Admin

The Super Admin has complete control over the entire platform and manages all registered organizations.

Responsibilities:

- Add, update, or remove organizations.
- Approve organization registrations.
- Create and manage organization staff counter accounts.
- Monitor system-wide analytics and reports.
- Configure AI prediction settings.
- Monitor database and cloud services.

- Manage user roles and permissions.
- Maintain overall system configuration.

2.4 Assumptions & Dependencies

Assumptions

The proposed system is developed based on the following assumptions:

- Users have access to smartphones, computers, or web browsers with an internet connection.
- Organizations maintain accurate service schedules and queue information.
- Cloud services remain available throughout system operation.
- Historical queue data is available or will be generated for AI model training.
- Users provide valid registration information.
- Service providers regularly update appointment availability.

Dependencies

The successful operation of the system depends on the following components:

- Stable internet connectivity.
- Cloud-hosted database (Azure SQL Database).
- ASP.NET Core Web API server.
- Flutter-supported platforms.
- AI prediction module developed using Python.
- Push notification service (Firebase Cloud Messaging).
- Optional CCTV or IoT devices for crowd monitoring.